

MANOHAR PATTANAYAK

AI Solution Security Architect | AI Security | Cloud Security | Application Security

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AI Solution Security Architect with nearly 3 years of experience designing secure architectures for AI-integrated, cloud-native, and enterprise applications across BFSI, Fintech, Insurance, and Healthcare. Specializes in AI security architecture, threat modeling, application security testing, and cloud security across AWS, Azure, and GCP.

EXPERIENCE

Consultant – AI & Solution Security Architect | NuSummit Cybersecurity | Mumbai

March 2026 – Present

- Conducted network architecture and firewall security reviews (segmentation, VPNs, IDS/IPS), identifying critical security gaps and reducing misconfigurations by 30% across client environments.
- Performed AI security architecture reviews for AI-integrated products, identifying prompt injection, model governance gaps, and LLM data privacy risks, reducing AI risk exposure by 35% across 5+ systems.
- Assessed cloud security architecture across AWS, Azure, and GCP, uncovering IAM misconfigurations and encryption gaps and achieving 100% remediation compliance for 8+ critical applications.
- Performed threat modeling for cloud infrastructure and LLM deployments, delivering prioritized remediation roadmaps that reduced risk exposure by 40% across critical workloads.

Senior Security Analyst – Application Security Architect | Deloitte Touche Tohmatsu India LLP

| Mumbai

August 2023 – March 2026

- Designed and reviewed application security architectures for 15+ enterprise clients across BFSI, Fintech, and Healthcare, reducing architectural risk exposure by 40%.
- Executed application security testing (SAST, DAST, API security) across 20+ web and mobile applications, uncovering OWASP Top 10 vulnerabilities and reducing critical findings by 45%.
- Performed AI security architecture reviews for AI/ML solutions, identifying model poisoning and adversarial input risks, delivering remediation roadmaps that reduced AI risk exposure by 35%.
- Executed compliance audits aligned with ISO 27001, PCI DSS, SOC 2 Type II, and RBI guidelines, achieving 100% audit clearance for all BFSI clients.

Security Analyst | SDI | Bhubaneswar

December 2022 – May 2023

- Performed application and network security testing across web, mobile, and client-server applications, uncovering 50+ vulnerabilities and reducing network risk exposure by 30%.
- Collaborated with development teams to validate security fixes, achieving zero re-occurrence of identified vulnerabilities across all retested systems.

PROJECTS

Security Audit – Payment Gateway System

- Conducted penetration testing across 100+ endpoints, identifying and mitigating 15+ high-severity vulnerabilities and reducing overall attack surface by 35% with zero re-occurrence during retesting.

ATM Security Audit – Logical & Physical Controls

- Assessed security controls for 50+ ATMs, achieving 100% RBI compliance and reducing fraud risk exposure by 20% through targeted mitigation recommendations.

SKILLS

AI & Architecture Security: AI Security Architecture, LLM Security, Prompt Injection, Model Governance, Threat Modeling, Secure SDLC, API Security

Application Security: SAST, DAST, OWASP Top 10, Penetration Testing, AppSec Testing, Architecture Reviews

Cloud Security: AWS, Azure, GCP, IAM, Cloud Compliance, Security Posture Management

GRC & Compliance: ISO 27001, ISO 42001, SOC 2 Type II, PCI DSS, RBI Framework, ITGC

Certifications: ISO 42001:2023 AI Management Lead Auditor | ISO 27001:2022 Lead Auditor | CEH Practical | Certified AppSec Practitioner

Programming: Python, Java, SQL, Bash/Linux Scripting

EDUCATION

B.Tech in Electronics & Communication Engineering | CGPA: 8.2 / 10 | Silicon Institute of Technology, Bhubaneswar | 2019 – 2023